

Nikhil Sathyanarayana Shastry

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EDUCATION

Master of Science, Computer Science
University of Southern California, Los Angeles, CA
Bachelor of Engineering, Computer Science
B.M.S College of Engineering, Bangalore, India

Aug 2023 - Present
GPA: **3.95**/4.00
Aug 2017 - Aug 2021
GPA: **3.8**/4.00

SKILLS

Programming Languages: Python, Java, C++, SQL, Javascript, Typescript, Golang
Web Technologies: Node.js, Express.js, React.js, HTML, CSS, Prisma, Bootstrap, Tailwind CSS
Tools/Cloud Technologies: Git, AWS, GCP, Docker, Kubernetes, Github Actions
Databases: MySQL, PostgreSQL, MongoDB, Redis

EXPERIENCE

Full Stack Engineer, Core Center of Excellence in Nano Imaging, Los Angeles, CA *Apr 2024 - Present*

- Re-engineered a legacy PHP website into a RESTful web application using React.js and Node.js, and transitioned it to Azure Web Apps for improved performance
- Developed an Electron.js application, automating instrument usage logging and simplifying the previously manual billing process, thereby increasing operational efficiency by 80%

Software Engineer, Bosch Global Software Technologies, Bangalore, India *Nov 2021 - Jul 2023*

- Developed an enterprise-level tool using Java to convert software requirements in ARXML format to reStructured text format
- Completed implementing the new tool across 5 departments, resulting in a 40% reduction in software requirement errors
- Increased the efficiency of the A2R tool by 30% and reduced the amount of time spent on maintenance by 60%, resulting in significant cost savings and productivity improvements
- Devised a custom python tool to automate the conversion of 100+ Confluence pages into RST format, enabling the creation of interactive web pages deploying Sphinx

Software Engineer Intern, Hewlett Packard Enterprise, Bangalore, India *Feb 2021 - Jul 2021*

- Developed a full stack web application using React.js, Node.js, and MongoDB to streamline the data archival process, reducing server access request times by 40% and improving project efficiency by 15%
- Implemented RESTful APIs and JWT authentication for secure communication between engineers and VPs, automating approval workflows and integrating with HPE's email system for notifications
- Created an SOS alert feature for critical application downtime, connecting to HPE's on-call engineer database and reducing response time for high-risk issues by 30%

PROJECTS

Stock Simulator Pro

- Designed and implemented a full-stack stock trading simulator as an academic project, utilizing React, Node.js, and Express, and integrating real-time market data from Finnhub and Polygon.io APIs to create a realistic trading environment
- Engineered an efficient server-side caching system and implemented lazy loading techniques, optimizing data retrieval and significantly enhancing the application's performance and responsiveness
- Developed interactive, zoomable stock charts using Highcharts library, incorporating various technical indicators to provide users with comprehensive market analysis tools and improve the educational value of the simulator
- Successfully deployed the application on AWS (EC2, S3, CloudFront), demonstrating proficiency in cloud services and DevOps practices, and ensuring high availability and scalability in a simulated production environment

Misinformation Analysis and Detection of Low-Credibility News

- Implemented a Machine Learning model for neural social bot detection on Twitter based on comprehensive analysis of current research papers, integrating user-profile, social, and language-based attributes
- Applied Natural Language Processing techniques for sentiment analysis and text classification to enhance low-credibility news detection, following best practices from recent literature
- Developed a data processing pipeline to handle Twitter data, adapting methodologies from published studies to manage and analyze social media content effectively
- Conducted feature selection and engineering based on findings from multiple research papers, identifying key attributes for bot detection and misinformation classification
- Technologies applied: Python, Scikit-learn, Pandas, NLTK for natural language processing, and Git for version control